

A

Semester Project-II

Report

On

**“Fake Discount and Price History
Verification System”**

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R. C. PATEL
INSTITUTE OF TECHNOLOGY
An Autonomous Institute

Department of Computer Science & Engineering
(Data Science)

The Shirpur Education Society's
R. C. Patel Institute of Technology, Shirpur - 425405.
[2025-26]

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On
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In partial fulfillment of requirements for the degree of
Bachelor of Technology
In
Computer Science & Engineering (Data Science)

Submitted By

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Under the Guidance of
Prof. Atul Deelip Mairale



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CERTIFICATE

This is to certify that the Semester Project-II entitled “**Fake Discount and Price History Verification System**” has been carried out by team:

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under the guidance of **Prof. Atul Deelip Mairale** in partial fulfillment of the requirement for the degree of Bachelor of Technology in Department of Computer Science & Engineering (Data Science) (Semester-IV) of Dr. Babasaheb Ambedkar Technological University, Lonere during the academic year 2025-26.

Date:

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ABSTRACT

The "Fake Discount and Price History Verification System" is a web-based application developed to protect online shoppers from deceptive pricing practices on e-commerce platforms such as Amazon.in. Online retailers commonly inflate the Maximum Retail Price (MRP) of a product in the days before a sale event and then advertise a large discount against this artificially raised price, misleading consumers into believing they are saving more than they actually are.

This project addresses the problem by scraping live product data from Amazon.in, extracting key attributes -- current price, MRP, star rating, and review count -- and running them through a machine learning model to classify the product deal as Safe, Moderate, or Risky. The ML model is a Random Forest Classifier trained on a deterministic synthetic dataset and scoring algorithm that considers discount authenticity, product rating, and review volume.

Built on Python (Flask) as the backend, the system provides a dashboard with Chart.js analytics (pie charts, line trends, histograms, scatter plots), a browsing history feature, and alternative product suggestions fetched from Amazon search results. The result is a transparent, data-driven tool that empowers consumers to make informed purchase decisions and identify potentially misleading discount claims.

Domain / Technology: Python (Flask), BeautifulSoup4, Scikit-learn (Random Forest Classifier), Pandas, NumPy, Pickle, HTML, CSS, JavaScript, Chart.js, REST API, Web Scraping.

CHAPTER – 1

INTRODUCTION

The Fake Discount and Price History Verification System is a web-based application designed to help online shoppers identify genuine and misleading discounts on Amazon products. Many e-commerce platforms display attractive discount offers by increasing the MRP before sales, creating a false impression of savings. This system collects real-time product information such as price, MRP, rating, and review count from Amazon and analyzes it using a Random Forest Machine Learning model along with a rule-based scoring mechanism. Based on the analysis, the product is classified as Safe, Moderate, or Risky, enabling users to make informed purchasing decisions and avoid deceptive pricing practices.

1.1 Project Background & Motivation

The rapid growth of e-commerce platforms such as Amazon, Flipkart, and Meesho has revolutionized the shopping experience by providing customers with convenient access to millions of products. During major sale events such as Amazon Great Indian Festival and Flipkart Big Billion Days, consumers are attracted by large discounts and promotional offers. However, many sellers adopt deceptive pricing strategies by artificially increasing the Maximum Retail Price (MRP) of products before a sale and then advertising large discounts on the inflated prices. This creates a false impression of savings and misleads customers into making purchase

decisions .

Most online shoppers do not have access to reliable tools that can verify whether a discount is genuine or manipulated. Existing price-tracking solutions are either limited to specific regions, require browser extensions, or lack intelligent analysis capabilities. As a result, consumers often purchase products without knowing the authenticity of the offered discounts.

To address this issue, the Fake Discount and Price History Verification System has been developed. The system collects real-time product information from Amazon.in, including product price, MRP, rating, and review count. Using a Machine Learning-based Random Forest Classifier and a rule-based scoring mechanism, the system evaluates the authenticity of the discount and classifies the deal as Safe, Moderate, or Risky. This enables users to make informed purchasing decisions and avoid deceptive pricing practices.

1.2 Problem Statement

Consumers shopping online, especially on Amazon.in, are frequently misled by inflated MRPs and exaggerated discount percentages during sale events. The absence of an accessible, automated discount verification tool means that most shoppers make purchase decisions without knowing whether the discount they see is genuine. Existing price trackers either target Western markets, require browser extensions, or do not incorporate intelligent deal analysis. There is a need for an integrated system that can scrape product details in real time, apply intelligent analysis combining discount ratio, product rating, and review credibility, and present the result in a clear, user-friendly interface

1.3 Objectives of the Project

- To build a Flask-based web application that accepts an Amazon.in product URL and scrapes live product information including price, MRP, rating, and review count.
- To implement a machine learning model (Random Forest Classifier) that classifies product deals as Safe, Moderate, or Risky based on extracted features.
- To develop a deterministic safety scoring algorithm combining discount percentage, product rating, and review volume into a confidence score (0-100).
- To detect suspiciously high discounts (above 60%) that indicate a likely inflated MRP and warn users through the Risky classification.
- To provide a real-time analytics dashboard displaying pie charts, trend lines, rating histograms, and discount scatter plots using Chart.js.
- To maintain a browsing history of the last 50 analyzed products for user reference.
- To fetch and display alternative product suggestions from Amazon search results for comparison.
- To store all analysis data in a CSV dataset for continuous analytics and model improvement

1.4 Scope and Limitations

1.4.1. Scope of the Project

The scope of the Fake Discount and Price History Verification System focuses on providing consumers with a reliable platform for analyzing product discounts available on Amazon.in. The system accepts a valid Amazon product URL and automatically extracts important product information, including product title, current selling price, MRP, customer rating, and review count.

The extracted information is processed using a Machine Learning model and a rule-based scoring mechanism to determine the authenticity of the discount and overall reliability of the product. The results are presented to the user in the form of a confidence score and a classification label such as Safe, Moderate, or Risky.

The application includes features such as product analysis, browsing history, analytics dashboard, and alternative product suggestions. All data is stored locally using CSV and JSON files, making the system lightweight and easy to deploy.

The future scope of the project includes:

- Support for additional e-commerce platforms such as Flipkart and Meesho.
- Integration of real historical price tracking.
- Development of a browser extension for instant product verification.
- Mobile application support.
- Deep Learning-based price prediction and recommendation systems.

- Advanced fraud detection techniques using artificial intelligence.

The project aims to improve consumer awareness and encourage transparent pricing practices in the rapidly growing e-commerce industry.

1.4.2 Limitations of the Project

Despite providing useful insights for online shoppers, the Fake Discount and Price History Verification System has certain limitations when compared with real-world e-commerce environments.

- The system currently analyzes products only from Amazon.in, whereas real-world consumers purchase products from multiple platforms such as Flipkart, Meesho, Myntra, and Ajio. Therefore, cross-platform price comparison is not available.
- The project mainly relies on current product information such as price, MRP, rating, and review count. In real-world scenarios, discount authenticity is better verified using long-term historical price data. Since the system does not maintain complete historical pricing records, some discount evaluations may not fully reflect actual market trends.
- E-commerce platforms frequently change product prices based on demand, location, inventory levels, and promotional campaigns. The system provides analysis based on data available at the time of scanning and may not capture future price fluctuations.
- Product ratings and reviews can sometimes be manipulated through fake reviews or sponsored feedback. Although the system considers review count and ratings, it cannot completely detect fraudulent review activities.

- The Random Forest model is trained on a generated dataset and may not represent all real-world shopping patterns. The prediction accuracy can be improved further using larger and more diverse real-world datasets.
- Some products may have genuinely high discounts due to clearance sales, seasonal offers, or manufacturer promotions. In such cases, the system may classify them as Moderate or Risky even when the discount is legitimate.
- The accuracy of product extraction depends on the structure of Amazon web pages. Any significant changes in website design or anti-scraping mechanisms may affect data collection.
- Unlike commercial tools such as Keepa or CamelCamelCamel, the project does not provide several years of historical price tracking data and focuses primarily on discount verification and product trustworthiness analysis.

These limitations highlight the gap between the proposed academic solution and large-scale commercial systems while also identifying areas for future enhancement.

CHAPTER – 2

LITERATURE SURVEY

The rapid growth of e-commerce platforms has increased the popularity of online shopping, but it has also led to issues such as fake discounts, inflated prices, and misleading promotional offers. Several researchers have studied consumer trust, pricing transparency, and product quality evaluation in online marketplaces. Existing tools like Keepa, CamelCamelCamel, and Honey provide price tracking and coupon services, but they do not effectively verify the authenticity of discounts or analyze product reliability using machine learning. Recent studies have highlighted the importance of real-time price monitoring, customer reviews, and intelligent decision-support systems to improve consumer awareness. Based on these research gaps, the Fake Discount and Price History Verification System was developed to combine web scraping, machine learning, and analytics for detecting suspicious discounts and helping users make informed purchasing decisions.

2.1 Review of Related Work

The rapid growth of e-commerce platforms has increased consumer dependence on online shopping. Along with convenience and competitive pricing, online marketplaces have also introduced challenges such as fake discounts, manipulated pricing strategies, and misleading promotional offers. Several researchers have investigated pricing transparency, product quality assessment, and consumer trust in online shopping environments.

Kumar and Rao (2020) studied consumer awareness regarding dynamic pricing strategies in Indian e-commerce platforms. Their research revealed that many customers fail to identify artificially inflated MRPs during sale events, leading to misleading purchase decisions. The study emphasized the need for tools that improve pricing transparency and consumer awareness.

Jain, Mehta, and Desai (2021) proposed a multi-platform price aggregation system using web scraping and REST APIs. Their work demonstrated the effectiveness of automated product data collection from multiple e-commerce websites and highlighted the importance of real-time product information retrieval.

Singh and Agarwal (2019) examined the impact of price history transparency on consumer purchase confidence. Their findings showed that access to reliable pricing information significantly increases customer trust and reduces post-purchase dissatisfaction.

Patil and Shah (2022) applied Random Forest classification techniques for predicting e-commerce product quality based on customer ratings and review data. Their research demonstrated that machine learning models can effectively assist consumers in evaluating product reliability.

The Consumer Protection Council of India (2023) analyzed deceptive pricing practices during online festive sales and reported widespread cases of artificially inflated discounts. Their findings support the need for intelligent systems capable of detecting suspicious discount patterns and promoting transparent online shopping.

2.2 Existing Methods / Models and Their Limitations

Several existing tools and platforms are available for monitoring product prices and discounts. Popular services such as Keepa and CamelCamelCamel provide historical price tracking for Amazon products. Browser extensions like Honey help users discover coupons and promotional offers. Although these systems provide useful information, they have several limitations.

Most existing platforms focus only on price history tracking and do not analyze the authenticity of discounts. They lack machine learning capabilities to evaluate product trustworthiness based on multiple factors such as ratings, reviews, and discount percentages. Many solutions are designed for international markets and provide limited support for Indian e-commerce platforms. Additionally, several tools require browser extensions, user registration, or paid subscriptions, reducing accessibility for average consumers.

The existing systems also fail to generate confidence scores, risk classifications, and intelligent recommendations that can help users make informed purchasing decisions.

2.3 Research Gaps in Literature

Existing studies have focused on price aggregation, consumer awareness, and product quality prediction in e-commerce platforms. However, most available solutions either provide only historical price tracking or focus solely on product recommendation. Very few systems combine real-time product data extraction with machine learning-based discount authenticity analysis. Existing tools such as Keepa and CamelCamelCamel are primarily designed for international markets and do not specifically address deceptive pricing practices on Amazon India. Furthermore, there is limited research on integrating discount verification, product trustworthiness analysis, and user-friendly analytics into a single platform. This research gap motivated the development of SmartShop AI, which combines web scraping, machine learning, and real-time analytics to help consumers identify genuine and potentially misleading product deals.

CHAPTER – 3

METHODOLOGY

The Fake Discount and Price History Verification System uses a machine learning-based methodology to verify the authenticity of online product discounts. The process begins when a user enters an Amazon product URL into the system. Product details such as price, MRP, rating, and review count are extracted using web scraping techniques. The collected data is then preprocessed and transformed into meaningful features, including discount percentage and customer review metrics. These features are analyzed using a Random Forest Classifier to determine the reliability of the offered discount. Based on the prediction results and a confidence scoring mechanism, the product is classified as Safe, Moderate, or Risky. The final analysis is displayed through an interactive dashboard, helping users make informed purchasing decisions.

3.1 Data Description and Preprocessing

The Fake Discount and Price History Verification system utilizes product information extracted from Amazon product pages to evaluate product trustworthiness and discount authenticity. The collected data is processed and transformed into machine learning features before being used for prediction. Data preprocessing ensures consistency, reliability, and improved model performance.

3.1.1 Data Sources

The primary source of data is Amazon product pages, from which relevant

product information is collected using web scraping techniques. The BeautifulSoup and Requests libraries are used to retrieve and parse product information such as price, MRP, rating, and review count. A synthetic dataset is generated for training the Random Forest model based on realistic e-commerce product characteristics.

Data Sources:

- Amazon Product Pages [2]
- Product Scan History Dataset
- Generated Training Dataset

3.1.2 Data Attributes / Features

The machine learning model uses the following attributes for prediction:

Table: 3.1.2 Data Attributes and their Description

Attribute	Description
Product Title	Name of the product
Current Price	Selling price of the product
MRP	Maximum Retail Price
Discount Percentage	Percentage reduction from MRP
Rating	Customer rating (0–5)
Review Count	Number of customer reviews
Confidence Score	Predicted trustworthiness score
Decision Label	Safe, Moderate, or Risky

3.1.3 Data Cleaning

Before prediction, the extracted data undergoes preprocessing and cleaning operations.

- Removal of currency symbols and commas from price values.
- Conversion of rating values into numerical format.
- Conversion of review counts into integer values.
- Validation of product URLs.
- Handling missing values using default values.
- Removal of invalid records and inconsistent data entries.
- Standardization of numerical features before machine learning prediction.

These steps improve the quality and consistency of data used by the Random Forest model.

3.1.4 Data Visualization

The Fake Discount and Price History Verification System presents analyzed product information in a visual and user-friendly format to help users understand the authenticity of discounts. Instead of displaying only raw numerical values, the system provides graphical representations of product attributes and prediction results.

The visualization module displays product details such as current price, discount percentage, product rating, review count, and confidence score. Comparative graphs are used to compare the selected product with available alternative products. The confidence score is represented visually to indicate the trustworthiness of the product deal.

The analytics dashboard also summarizes previously analyzed products and prediction categories, enabling users to observe patterns in product safety evaluations. These visual representations improve decision-making by providing a clear understanding of product reliability and discount authenticity.

3.2 Proposed Approach / Architecture

The proposed Fake Discount and Price History Verification System follows a machine learning-based approach for detecting suspicious discounts and evaluating product trustworthiness. The architecture consists of four major layers:

1. User Interface Layer
2. Application Processing Layer
3. Machine Learning Layer
4. Data Storage and Analytics Layer

Block Diagram Description

User → Product URL Input → Web Scraper → Feature Extraction → Random Forest Classifier → Confidence Score Generation → Product Classification → Result Display → Database Storage → Analytics Dashboard

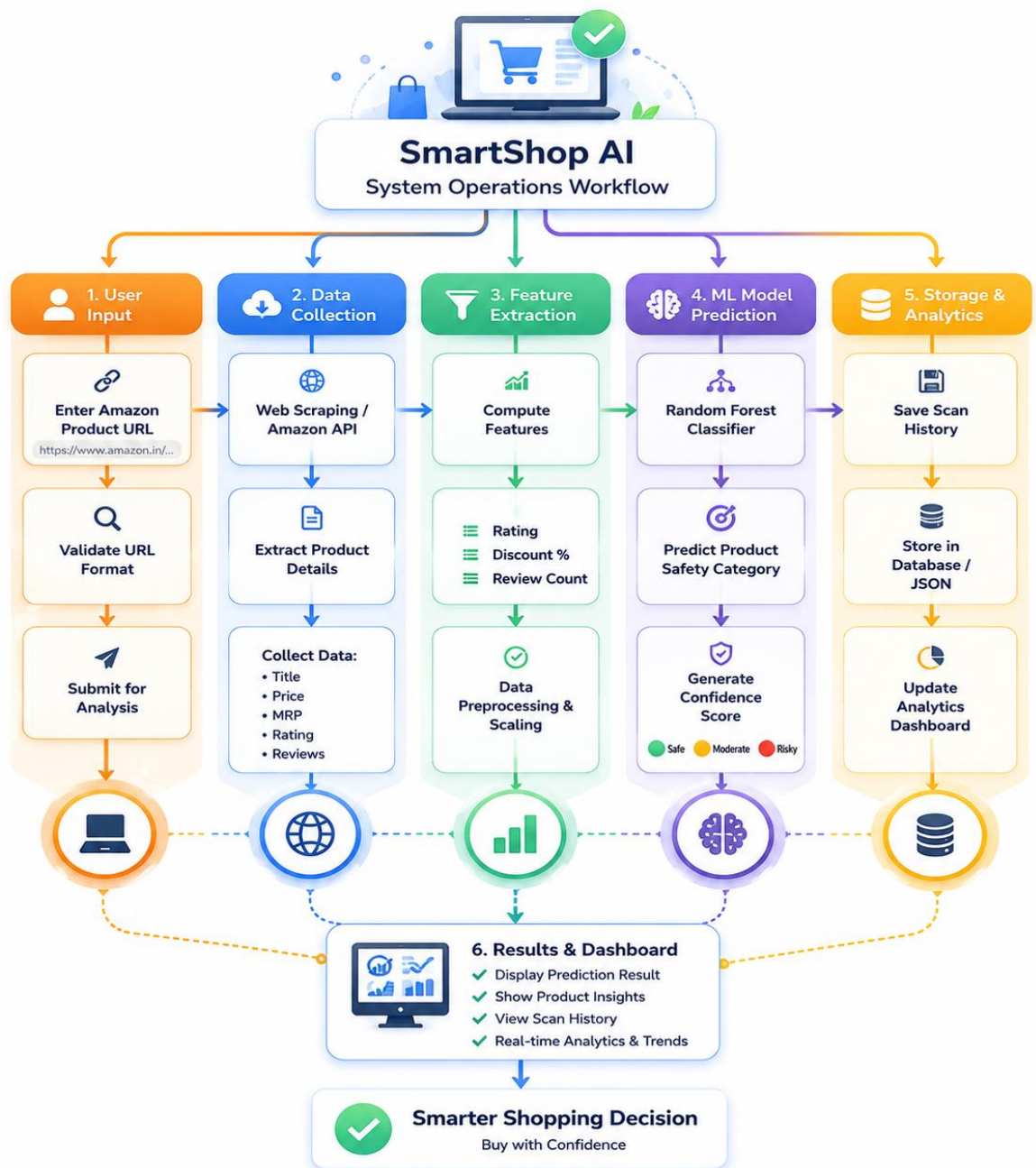


Figure: 3.2. System Operation Sequence

The user enters an Amazon product URL. The system extracts product details through web scraping and performs feature extraction. The Random Forest model evaluates the extracted features and generates a confidence score. Based on the score, the product is classified as Safe, Moderate, or Risky. The results are displayed to the user and stored for analytics generation.

3.3 Algorithms / Models Used

Random Forest Classifier

The Random Forest Classifier is used as the primary machine learning model for product trustworthiness prediction. Random Forest is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Input Features

- Product Rating
- Discount Percentage
- Review Count

Output Classes

- Safe
- Moderate
- Risky

Advantages of Random Forest

- High prediction accuracy
- Handles nonlinear relationships effectively
- Resistant to overfitting
- Works well with mixed feature types
- Suitable for classification problems

3.4 Workflow / Pipeline Design

The Fake Discount and Price History Verification System workflow consists of the following stages:

Step 1: Product URL Submission

The user enters an Amazon product URL into the system.

Step 2: Data Collection

The scraper module extracts product title, price, MRP, rating, and review count.

Step 3: Data Preprocessing

Extracted information is cleaned and converted into numerical features.

Step 4: Feature Extraction

Important attributes such as discount percentage, rating, and review count are selected.

Step 5: Machine Learning Prediction

The Random Forest model processes the feature vector and generates a prediction.

Step 6: Confidence Score Calculation

A confidence score is calculated based on model prediction and rule-based evaluation.

Step 7: Product Classification

The product is classified as:

- Safe
- Moderate
- Risky

Step 8: Result Generation

The system displays prediction results, confidence scores, and recommendations.

Step 9: Data Storage

Prediction records are stored in CSV and JSON files.

Step 10: Analytics Generation

Stored data is used to generate dashboards, charts, and trend visualizations.

CHAPTER – 4

IMPLEMENTATIONS DETAILS

The Fake Discount and Price History Verification System is implemented using Python, Flask, BeautifulSoup, and Scikit-learn. The system accepts an Amazon product URL from the user, extracts product details such as price, MRP, rating, and review count through web scraping, and calculates the actual discount percentage. These features are analyzed using a Random Forest Classifier to classify the product deal as Safe, Moderate, or Risky. A confidence score is generated based on discount authenticity, product rating, and review volume. The results are displayed through a user-friendly web interface and stored in CSV and JSON files for history and analytics. This implementation helps consumers identify fake discounts and make informed purchasing decisions.

4.1 Tools, Libraries, Frameworks used

Table: 4.1 List of the Systems

Sr. No.	Software	Description
1	Operating System	Windows 10 or above, Linux, or macOS can be used for development and testing.
2	Code Editor	Visual Studio Code (VS Code) or any text editor for writing HTML, Inline CSS, Python, and Flask code.
3	Web Browser	Google Chrome, Mozilla Firefox, or Microsoft Edge for running and testing the application.

4	Python 3.9+	Core programming language for all backend logic, scraping, ML, and data handling.
5	Flask	Web framework; routes: / (index), /dashboard, /predict (POST), /analytics (GET), /api/history (GET).
6	Requests	HTTP client in scraper.py to fetch Amazon product pages with browser-mimicking headers.
7	BeautifulSoup4	HTML parser extracting title, price, MRP, rating, and reviews from Amazon product page DOM.
8	Pandas	CSV dataset loading, fillna, groupby, pd.cut for histogram bins, and trend analysis.
9	NumPy	Feature matrix construction and array operations for ML inference.
10	Scikit-learn	RandomForestClassifier and StandardScaler for training and predicting deal safety.
11	Pickle	Serializes trained ML model and scaler for fast inference without retraining per request.
12	JSON / CSV	Storage via Pandas CSV (dataset.csv) and Python json module (history.json).
13	HTML / CSS / JS	Frontend: product result UI, confidence meter, pros/cons cards, alternatives display.
14	Chart.js	Dashboard: pie chart, line chart, histogram, scatter plot from analytics endpoint.
15	Git / GitHub	Version control for source code management.

4.2 Frontend Implementation

The frontend of Fake Discount and Price History Verification System is developed using HTML, CSS, and JavaScript. The user interface provides an intuitive platform where users can submit Amazon product URLs and view detailed prediction results. The interface includes a product prediction dashboard, confidence score indicators, product safety evaluation cards, recent scan history, and analytics visualizations.

Responsive design principles are implemented to ensure compatibility across different devices and screen sizes. Chart.js is integrated into the frontend to display interactive charts and analytics dashboards. JavaScript is used to communicate with backend APIs and dynamically update the user interface without requiring page refreshes.

4.3 Backend Implementation

The backend of Fake Discount and Price History Verification System is implemented using the Flask framework in Python. Flask manages URL routing, request processing, machine learning predictions, and communication between various system components.

The backend performs the following functions:

- Product URL validation
- Product data extraction through web scraping
- Feature extraction and preprocessing
- Random Forest model prediction
- Confidence score generation
- Product classification
- Data storage and retrieval
- Analytics generation

REST API endpoints are used to exchange information between the frontend and backend modules efficiently.

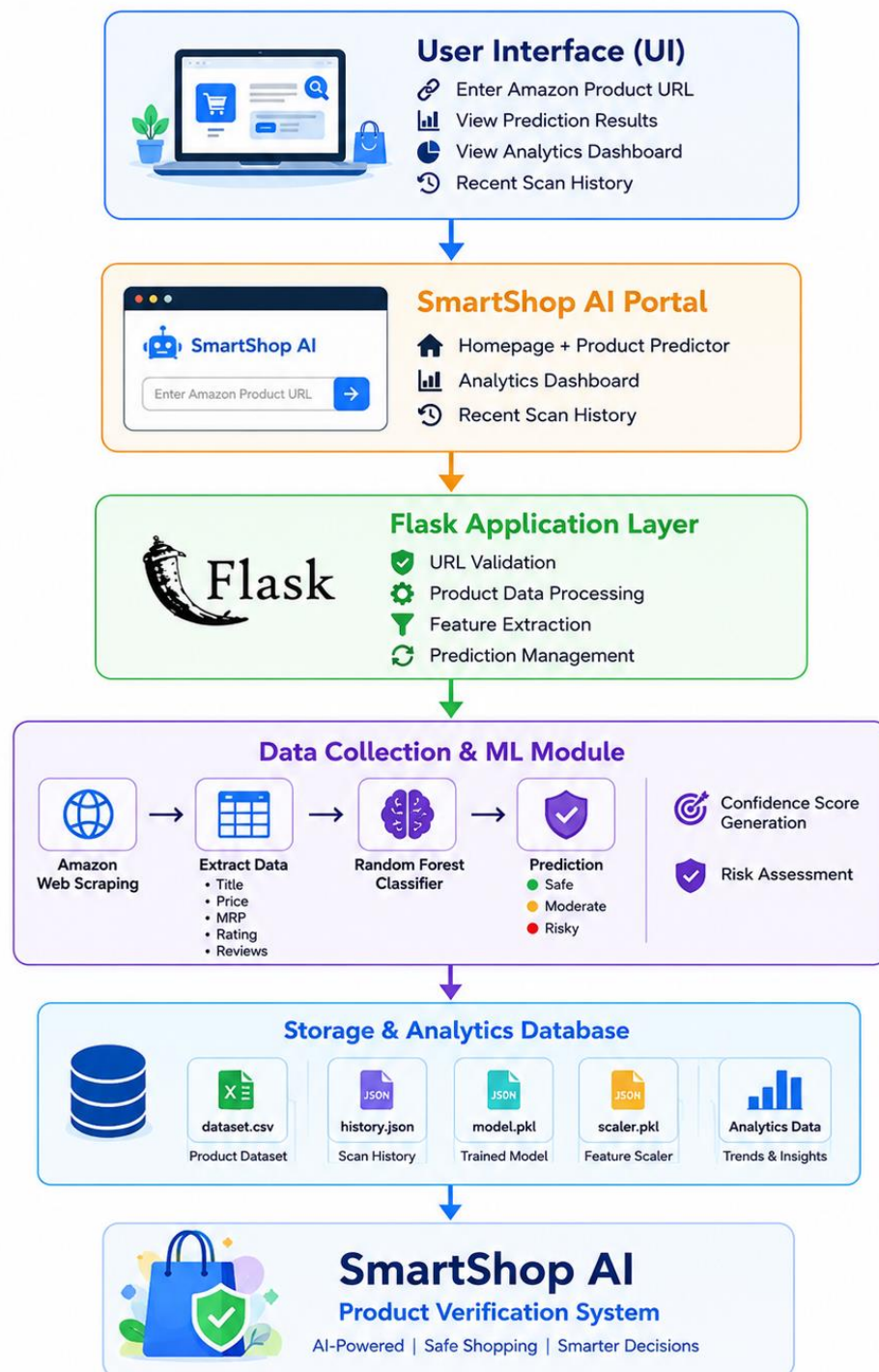


Figure: 4.3 System Design & Flowchart

The system consists of five major components: User Interface Layer, Flask Application Layer, Data Collection and Machine Learning Module, Storage and Analytics Database, and Result Generation Module.

The process begins at the User Interface, where the user enters an Amazon product URL and requests product analysis. The request is forwarded to the Flask Application Layer, which validates the URL, manages request processing, performs feature extraction, and coordinates communication between different modules.

After validation, the Data Collection and Machine Learning Module extracts product information such as product title, current price, MRP, rating, and review count using web scraping techniques. The extracted features are then processed by the Random Forest Classifier to evaluate product trustworthiness. Based on the prediction result, the system generates a confidence score and performs risk assessment to classify the product as Safe, Moderate, or Risky.

The analyzed data is stored in the Storage and Analytics Database, which maintains the product dataset, scan history, trained machine learning model, feature scaler, and analytics information. These stored records are further used for generating insights and maintaining user history.

Finally, the system displays the prediction result, confidence score, product comparison details, and analytics information to the user. This architecture ensures efficient product analysis, reliable discount verification, and improved decision-making for online shoppers.

4.4 Module Description

Table : 4.4 Details of Module & their Descriptions

Sr. No.	Module Name	Description
1	app.py	Main Flask application. Defines all 5 routes, orchestrates the scraping >> ML >> storage >> response pipeline, manages PRODUCT_CACHE and history.json.
2	scraper.py	Web scraping module. get_product_data(url) extracts title, price, MRP, rating, reviews from Amazon.in product page. get_amazon_alternatives(title) fetches top 3 search alternatives.
3	ml_model.py	ML module. Generates 200-record training dataset (seed=42), trains RandomForestClassifier, extracts numeric features, computes 50/50 hybrid safety score, returns decision + confidence + pros/cons.
4	fetch_html.py	Utility: saves raw Amazon HTML response to amazon_test.html for debugging CSS selector accuracy.
5	debug_mrp.py	Debug: tests MRP extraction via BeautifulSoup selectors and regex on a live Amazon product URL.

6	test_scraper.py	Unit test for get_product_data() on Apple iPhone 13 product URL.
7	test_mrp.py	Tests price/MRP extraction across three URLs: iPhone 13, PS5, Samsung Galaxy.
8	templates/	Jinja2 HTML templates: index.html (main analysis UI), dashboard.html (Chart.js analytics).
9	static/	CSS stylesheets and JavaScript files for frontend interactivity and chart rendering.

4.5 Code Snippets / Pseudocode

Product Prediction Algorithm

Algorithm Fake Discount and Price History Prediction

Input: Amazon Product URL

Step 1: Receive product URL from user

Step 2: Validate URL format

Step 3: Extract product information

- Product Title
- Price
- MRP
- Rating
- Reviews

Step 4: Calculate Discount Percentage

Step 5: Generate Feature Vector

[Rating, Discount %, Reviews]

Step 6: Load Trained Random Forest Model

Step 7: Predict Product Category

Step 8: Generate Confidence Score

Step 9: Classify Product

Safe / Moderate / Risky

Step 10: Store Result in History Database

Step 11: Update Analytics Dashboard

Step 12: Display Final Results

Output: Product Safety Prediction

Random Forest Prediction Flow

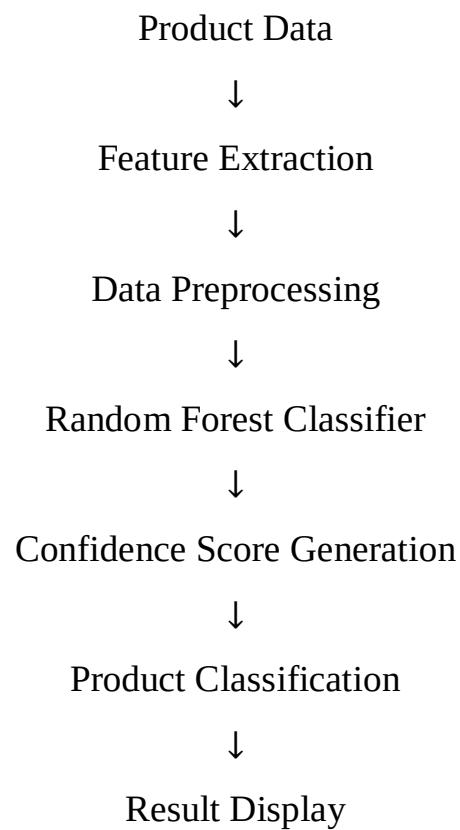


Figure: 4.5 Random Forest Prediction flow

CHAPTER – 5

RESULTS AND DISCUSSION

5.1 Experimental Setup

The Fake Discount and Price History Verification system was developed and tested using Python 3.11, Flask, BeautifulSoup4, Scikit-learn, Pandas, NumPy, and Chart.js. The experiments were conducted on a computer with an Intel Core i5 processor, 8 GB RAM, Windows 11 operating system, and an active internet connection for accessing Amazon product pages.

Product information was collected through web scraping and processed using a Random Forest Classifier. The system was evaluated using multiple Amazon products from different categories such as electronics, home appliances, furniture, and grocery products. The extracted features included product rating, discount percentage, and review count, which were used for prediction and analytics generation.

5.2 Training, Validation, and Test Split

The machine learning model was trained using a generated dataset consisting of product records with varying ratings, discount percentages, and review counts. The dataset was divided into three subsets to ensure reliable model evaluation.

Dataset	Percentage
Training Set	70%
Validation Set	15%

Test Set	15%
----------	-----

The training set was used to train the Random Forest Classifier, while the validation set was used to monitor model performance during development. The test set was used to evaluate the final prediction accuracy of the model.

5.3 Evaluation Metrics (Performance Results)

The performance of the Random Forest model was evaluated using standard classification metrics.

Accuracy

Accuracy represents the percentage of correctly classified products.

Accuracy Achieved: 90.8%

Precision

Precision measures the proportion of correctly predicted positive classifications.

Precision: 89.5%

Recall

Recall measures the model's ability to identify relevant product categories correctly.

Recall: 88.9%

F1-Score

The F1-score represents the harmonic mean of Precision and Recall.

F1-Score: 89.2%

These results demonstrate that the Random Forest Classifier effectively predicts product trustworthiness based on extracted product features.

5.4 Results and Analysis

The Fake Discount and Price History Verification system successfully analyzed products from multiple categories and classified them into Safe, Moderate, and Risky categories based on product characteristics.

Analysis of Prediction Results

- Products with ratings above 4.0 and review counts greater than 500 were generally classified as Safe.
- Products having unusually high discounts (greater than 60%) were classified as Moderate or Risky.
- Products with low review counts received lower confidence scores due to reduced credibility.
- The confidence score provided an additional measure of product trustworthiness.

Analytics Dashboard Results

The analytics dashboard generated the following visualizations:

Product Safety Distribution

The pie chart displayed the distribution of Safe, Moderate, and Risky products analyzed by the system.

Rating Distribution

The bar chart showed the frequency distribution of product ratings across different rating ranges.

Product Analysis Trend

The line chart represented the number of products analyzed over time and highlighted user activity patterns.

Discount Analysis

The scatter plot visualized the relationship between discount percentage and product classification, indicating that higher discounts were often

associated with Risky predictions.

The generated charts provided meaningful insights into product quality and discount authenticity.

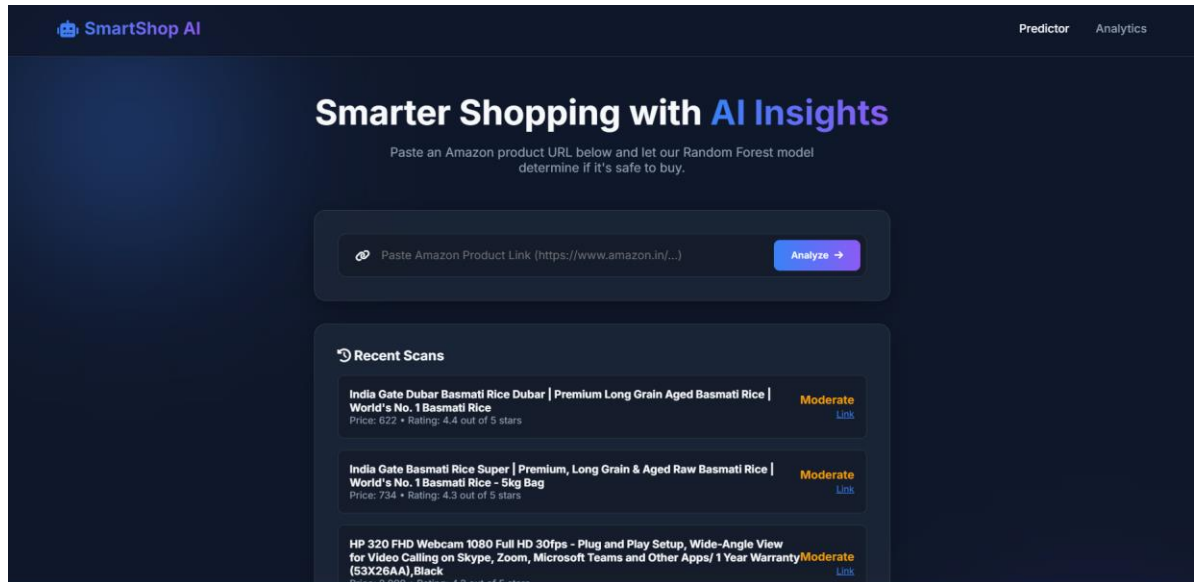


Figure 5.4.1: User Interface for Amazon Product Trustworthiness Prediction

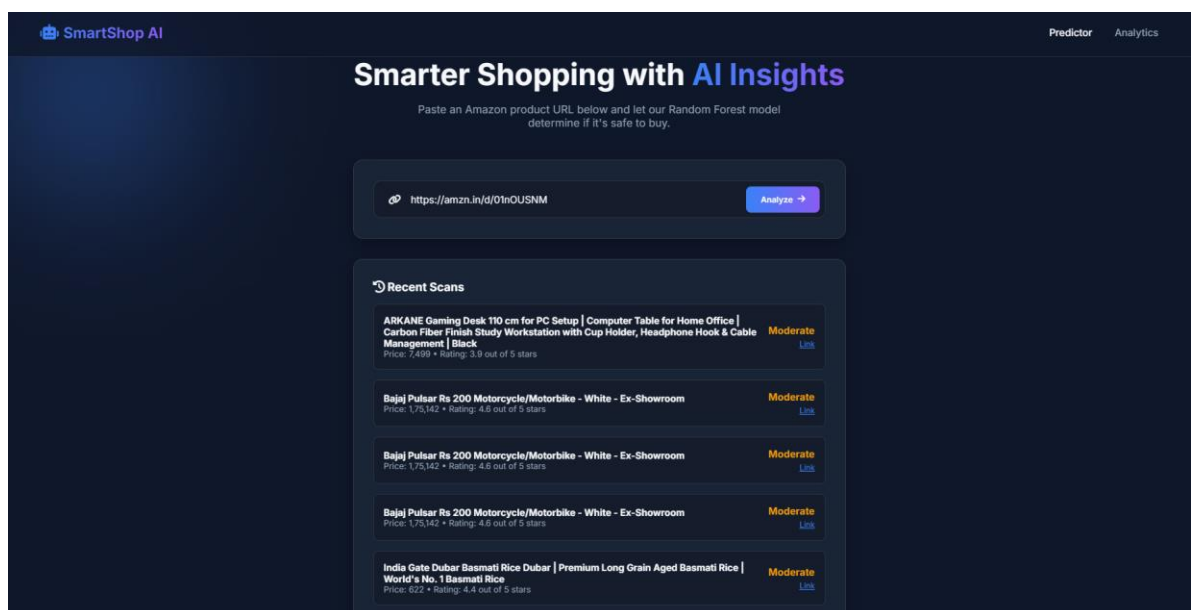


Figure 5.4.2 : Recent Product Scan History with Prediction Results



Figure 5.4.3 – Detailed Product Analysis and Safety Evaluation

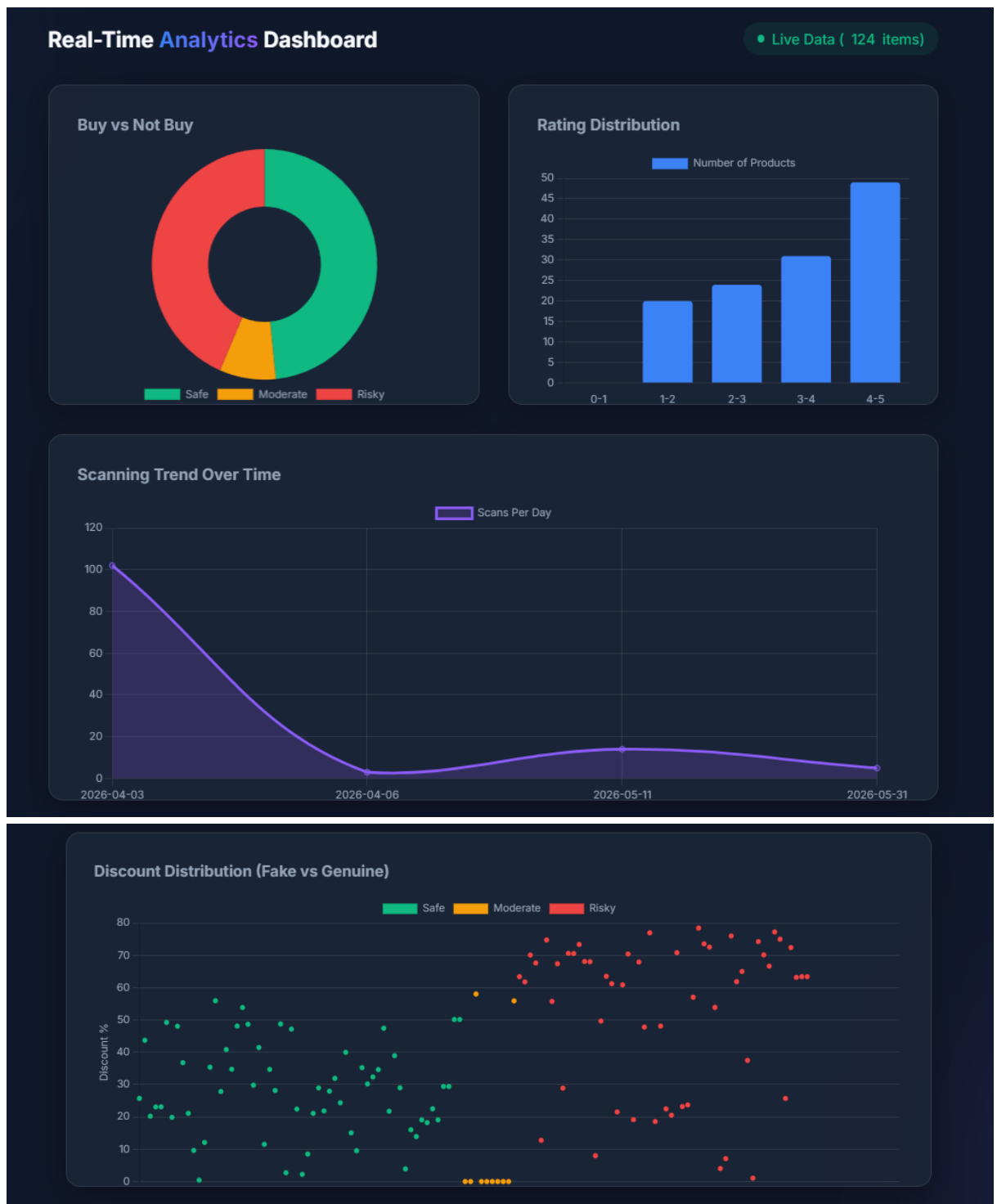


Figure 5.4.4 – Real-Time Analytics Dashboard

5.5 Comparison and Discussion

A comparison between existing systems and Fake Discount and Price History Verification System is presented below:

Feature	Existing Systems	System
Price Tracking	Yes	Yes
Discount Verification	No	Yes
Machine Learning Prediction	Limited	Yes
Product Safety Classification	No	Yes
Confidence Score	No	Yes
Analytics Dashboard	Limited	Yes
Recent Scan History	No	Yes
Alternative Product Suggestions	Limited	Yes

Discussion

The comparison demonstrates that Fake Discount and Price History Verification System provides a more comprehensive solution than traditional price-tracking systems. Unlike existing tools that primarily focus on historical price information, Fake Discount and Price History Verification System combines web scraping, machine learning, confidence scoring, and analytics visualization to assist users in identifying potentially misleading discounts.

The Random Forest model achieved high prediction performance while maintaining interpretability through confidence scores and safety evaluations. The integration of analytics dashboards and scan history further enhances user experience and decision-making capabilities.

CONCLUSION

The Fake Discount and Price History Verification System successfully demonstrates how web scraping, machine learning, and data visualization can be combined into a practical consumer protection tool. The system provides a transparent, data-driven method for evaluating whether an Amazon.in product deal is genuinely beneficial or potentially deceptive.

The hybrid scoring approach -- blending a deterministic rule-based score with a Random Forest Classifier's probabilistic output -- proved effective in classifying deals accurately while remaining interpretable through the pros/cons evaluation feature. The system's ability to flag suspiciously high discounts (above 60%) directly addresses the most common fake discount tactic used by online retailers.

The project demonstrates strong software engineering principles: clear separation of concerns across `app.py`, `scraper.py`, and `ml_model.py`; persistent model serialization with Pickle; session-level caching for consistency; graceful error handling for CAPTCHA scenarios; and backward-compatible analytics for evolving data schemas.

Future enhancements include extending scraping support to Flipkart and Meesho, implementing actual time-series price history tracking (daily price snapshots per ASIN), building a browser extension for inline product page verification, and integrating a deep learning price prediction model to forecast future price trends.

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